Analysis on Germany's Gas, Oil and electricity consumption

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***Abstract*—A time series analysis is conducted on electricity availability, inland natural gas consumption and oil and petroleum products deliveries in Germany to assess the economy of Germany and the changes in the energy sector of Germany. We model time series spanning from 2008-2023 using seasonal adjustments and ARIMA models. We assess the predictive ability of the models of the time series. Overall the predictions of electricity availability and oil and petroleum products deliveries follow the expected shape of the data. However, inland natural gas consumption is predicted to not increase as much as it has in previous years. Germany is predicted to be stable economically and have few changes in the energy sector.**

I. Introduction

The demand for electricity, as a fundamental energy resource, and the consumption of inland natural gas is intricately tied to the economic, social, and environmental development of countries. Understanding the patterns and fluctuations in electricity production is vital for many people, including politicians, researchers and the public.

In recent years Germany has had an energy transition plan, known as the Energiewende[1]. This transition aims to significantly reduce greenhouse gas emissions and dependence on nuclear power while fostering a sustainable energy future. Central to this transition is the integration of renewable energy sources, such as wind, solar, and biomass, into the electricity grid and the phasing out of non-renewable energy sources, such as oil and natural gas. Amidst this transition, gas consumption patterns reflect the changes in economic fluctuations, technological advancements, policy interventions, and societal preferences.

The dynamics of oil and petroleum product deliveries also serve as an important indicator for understanding the economic, industrial and environmental landscapes of a nation. However, the significance of understanding oil and petroleum product deliveries in Germany extends beyond just economic indicators. Due to the national atmosphere surrounding climate change and sustainable development, the patterns observed in energy consumption and delivery are pivotal in assessing progress toward renewable energy adoption, emissions reduction targets, and the transition to a greener economy.

This report embarks on a comprehensive analysis of the time series data pertaining to inland gas consumption, electricity availability and oil and petroleum product deliveries in Germany. We aim to unearth the trends, seasonal variations and log-term trajectories whilst unraveling the complex interplay of variables that influence the energy landscape in Europe's largest economy[2].

We perform exploratory analysis of the data by looking at the plot of the data, the ACF of the data and the PACF of the data to discover any seasonal components in the data. We then analyse why any seasonal components may appear in each time series.

We check stationarity using the Augmented Dickey Fuller(ADF) test and difference the data until it shows stationarity. We decide on what model fits the data when we look at the output of the ADF and the PACF of the data. We test a variety of ARIMA models to improve the models accuracy by comparing the Akaike Information Criteria(AIC) of similar ARIMA models.

To illustrate the predictive utility of our model, we forecast the first three months of 2024 for all sets of data. The data is split into training and test data so that the accuracy of the forecast can be evaluated and adjusted prior to the prediction of the first three months of 2024.

The outline of the paper is as follows. A description of the data is given in Section II as well as the methodology used. Section III presents the models selected and the results of the forecasting. The report concludes with a brief overview in Section IV.

Elementary knowledge of basic time series definitions is assumed.

II. Methods

A. Data & Preliminary Analysis

Our data are Oil, Gas and electricity consumption based in Germany. Fist of all, we need to upload our data set onto R. In general, there are 10 variables in each data set, in order to make our analysis easier, we are going to narrow our variables down to 2, which are time and consumption. Also, we will name these 2 restricted variables differently base on different data set, in general, it will be time and the unit. After this, we plot our data out as a linear line with X-axis names time and Y-axis names the designated name.

This is gas consumption data showing in Fig1. By looking at the plot on the left, it shows a decreasing trend in the long term. Also, there could be a seasonality in the data by just looking at the plot on the left. We then look at their ACF and PACF graph to make sure if there are any seasonality in the data. By looking at the ACF graph for gas consumption in the middle, it shows a clearly seasonal trend as there is a repeat pattern every 12 months. This is reasonable as people tend to use the same amount of gas every year. Both plot and ACF graph indicates that there are some peaks and bottoms in the data, this is because people tends to use more gas in the

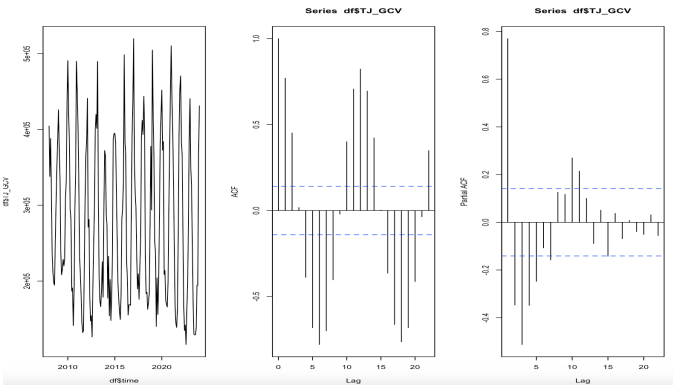


Fig. 1. Gas consumption

winter than summer. By analyzing and accounting for trend and seasonality, we can better interpret the historical patterns in the data and build models that capture these components.

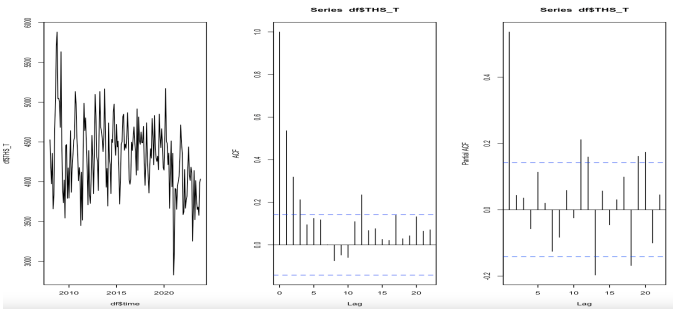


Fig. 2. Oil consumption

Fig2 is the exploratory data analysis for oil in Germany. It is the same to fig1, the plot on the left is the linear line that include every data. It doesn't look like the linear line for gas at all, there's no significant patterns that can be spotted by eyes. It is then confirmed by ACF graph, there is no repeating pattern showing in the graph, which indicates that the oil data does not have a seasonal trend. If we look at our plot on the left, we can notice that there are some outliers in it, such as the huge increase around 2009 and the sudden drop around 2021. In context, the huge drop might be caused by COVID-19, because not many people were able to use any transport, people got locked in their home, this will have a huge impact on the demand of oil in Germany. Therefore, when we are fitting our model into the data and making prediction, we might find that our model will not be able to fit our observed data perfectly, this will be caused by the outliers. So we can exclude the outliers when we fitting our model.

Finally, fig3 is the exploratory analysis for electricity in Germany. In general, the plot on the left shows there is a long-term decreasing trend within the data, also, it seems have a seasonality within the data set. It is the confirmed by the ACF graph in the middle, there is clearly a repeated patterns which happens every 12 months, we can also find out there are peaks in each pattern, this is probability that people tends

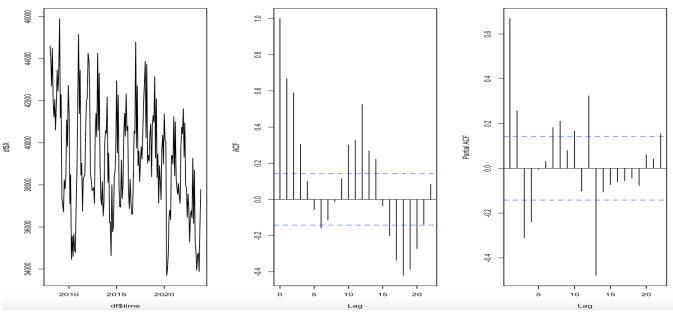


Fig. 3. Oil consumption

to use more electricity in winter than summer. After all 3 exploratory analysis purely based on the data itself, none of them are white noise process, because there are spikes pass over the blue boundaries in any ACF and PACF graph.

B. Model Fitting

We will only be using 80%-90% of data to do the model fitting which will be named as training data; and the remaining data which is test data will be used to test whether our model is a good model for forecasting.

We need to check if our data is stationary by doing Dickey-Fuller(ADF) test. By running the test in R, we need to check whether our data is smaller than 0.1, it is a stationary data if it is smaller than 0.1, and not if it is bigger. If we find out that our data is not stationary, we can progress to difference our data, after that, we run Dickey-Fuller test again until we get our desired P value. Stationary data tends to be more predictable, making it easier to build reliable and accurate models.

We then look at our ACF and PACF graph from our original data to decide which model we fit into ARIMA function. There are 2 situations, cuts off and tails off, Cuts off means that it becomes zero abruptly, and tails off means that it decays to zero asymptotically (usually exponentially). We look at the ACF and PACF graph with cuts off anf tails off in head. We use AR(p) model if it is tails off in ACF and cuts off in PACF after lag(q); on the other hand, if we have cust off in ACF and tails off in PACF after lag(p), we then use MA(p) model. More over, we will use ARIMA(p,q) if lag(p) and lag(q) in ACF and PACF are both tails off.

After that, we can check if our chosen model's residuals forms a white noise process, this is a way to validate our ARIMA model stands, there is no significant biased in the model. This is a crucial step in time series analysis to ensure that the chosen model adequately captures the underlying patterns in the data and that the model's predictions are reliable. We can now to improve our model's accuracy by restricting our coefficients, we can set any coefficients that are close to 0 into zero to make our model more accurate. IN order to check our restricted model is better than the original model, we need to compare their AIC value, we should expect to get a smaller AIC value from the restricted model compare to the original model. Finally, we check the ACF and PACF

on the residuals of the restricted model to see if the model forms a white noise process.

C. Forecasting

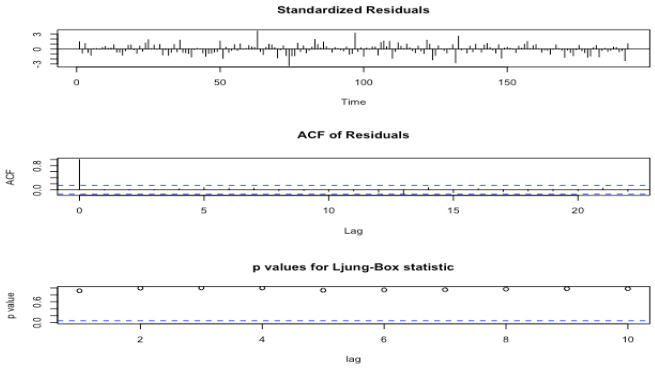
The main task here is to predict the upcoming 3 months values. Here we will use another package called forecast, and using the command called predict in the package to perform the forecasting for the next 3 months. Before that , we need to use our model to generate the same amount of data that matches with our test data. Plotting them out on the same graph, our predicted test data should have the same pattern as our observed data, and everything should be in the 95% confidence interval. This tells that our model is good fit and the predictions will be reasonable.

After that, we will use our model to forecast the next 3 months value and plot them out on the same graph with 95% confidence interval showing. We will then evaluate our forecasting by update our values and test that with fixed origin and rolling origin. The non-rolling origin approach evaluates the model over a fixed time period, providing consistency in the evaluation window. The rolling origin approach simulates a real-time forecasting scenario where the model is trained using data available up to a certain point in time and is then tested on the subsequent observations.

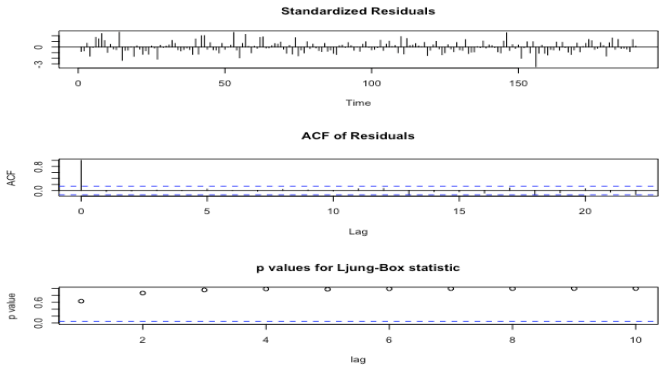
III. Results

A. Model Selection

In selecting a model for forecasting gas, oil, and electricity consumption in Germany we decided to use the ARIMA model. The reason we chose it is because the ARIMA model is great at managing time series data, flexibility in adapting to seasonal and non-stationary trends.
Lets start with gas. We decided to use ARIMA(12,0,0) model, because is particularly suited for seasonal time series with clear, consistent annual patterns.

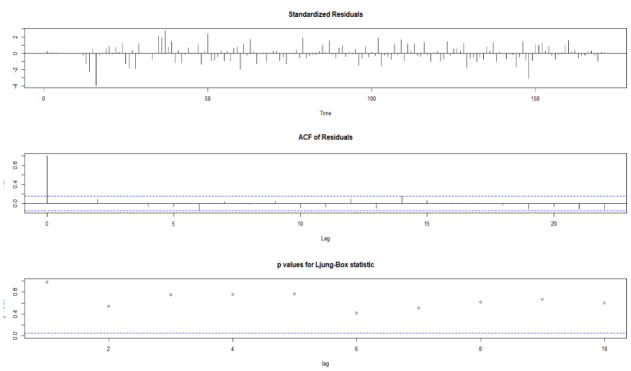


The plot indicates that there is no lag of the residuals and are within the confidence bands, implying that there is no significant lag left unexplained by the model. This suggests that the ARIMA(12,0,0) model has captured the time series lag effectively. the p-values for the Ljung-Box test at different lags All the p-values are high and well above the 0.05 threshold, indicating that there is no significant lag in the residuals. This is positive indicator that the model fits the data well..
For the modeling of oil data, we opted for the ARIMA(0,0,12) configuration due to having significant lags in the original dataset.



The ARIMA(0,0,12) model appears to be an appropriate fit for the oil time series data, as evidenced by the absence of significant lags in the residuals. Furthermore, the Ljung-Box test shows that the p-values are greater than the 0.05 suggesting it is a white noise process indicating a good model fit.

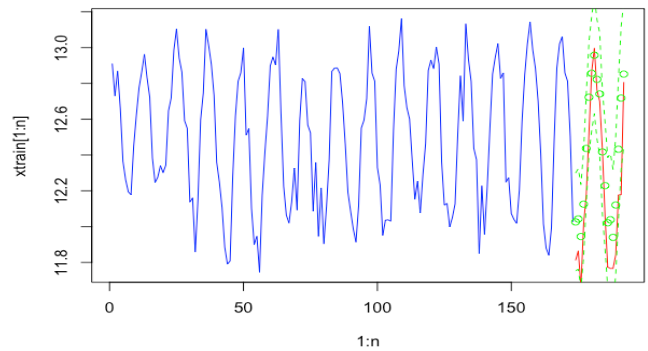
The selected ARIMA(0,1,1) model transformed with xlog for the electricity data demonstrates a reasonable fit.



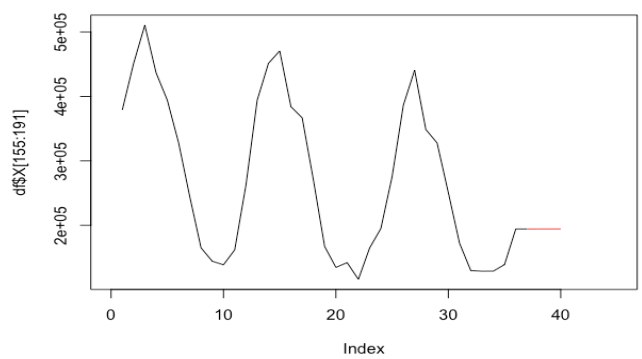
the ARIMA(0,1,1) model is good due to its simplicity and the minimal lag. The ACF shows that the model has captured the data's properties without lany lags that can be significant. This is further reaffirmed by the Ljung-Box test, where the p-values remain well above the 0.05 significance level, suggesting the residuals are independent and distributed randomly, meaning that the residuals behave like a white noise process.

B. Forecasting

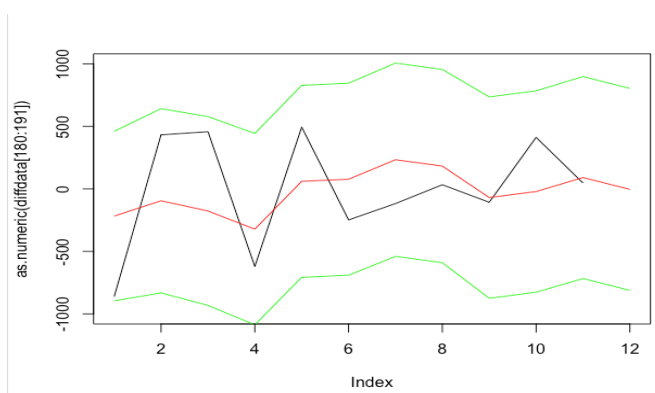
we will use our model to forecast the next 3 months value aiming to provide insights and predictions into Germany's Oil, Gas and Power economy.



for gas we used ARIMA(12,0,0) with a seasonal component projecting 19 time points into the future. the historical data is represented by a blue line, the test data is denoted by a red line, The forecasted values emerge as a green points and the prediction intervals are illustrated by dashed green lines. the predicted data continues with the trend of decreasing the gas consumption with time each year and because it is inside the confidence interval.

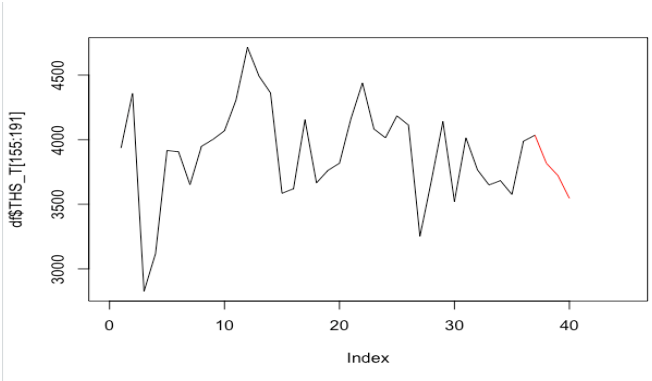


This graph represents the predicted values for the next 3 months. The historical gas consumption data, represented by a black line and red line that represents the predicted values. The red line's trajectory, while it seems simplistic in its incremental progression. It doesn't continue the trend of the historical gas consumption data.

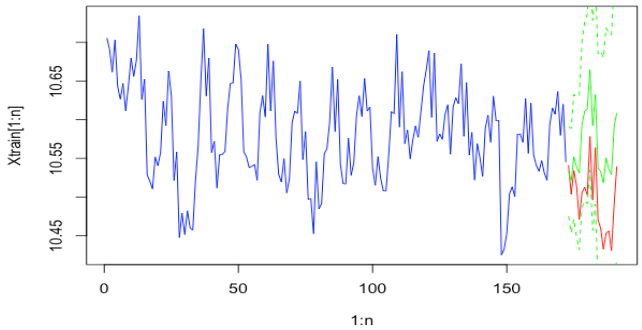


For oil we used ARIMA(0,0,12). The central black line represents the actual observed values, and the red line indicates

the forecasted values. The green lines seem to illustrate the upper and lower bounds of the forecast's confidence intervals. the model predicts a somewhat stable or slightly decreasing trend in the short term.



This graph represents the predicted values for the next 3 months. The black line represents The historical oil products delivery data, and the red line is the predicted values for the months of january february and march. As we can see the forecasted data follows the trend from the historical data that is a decrease in demand.



For electricity we used ARIMA(0,1,1) the historical data is represented by a blue line, the test data is denoted by a red line, The forecasted values emerge as a green points and the prediction intervals are illustrated by dashed green lines. The model appears to anticipate a continuation of the trend in the short term by decreasing electricity availability over time , but by the widening confidence bands, showing the inherent uncertainty in predicting further into the future.

IV. Discussion

This report has presented an analysis of the three time series of electricity availability, inland natural gas consumption and oil and petroleum products deliveries in Germany. The forecasts on the time series shows that for electricity availability and oil and petroleum products deliveries the predictions follow the expected shape of the data, however the inland natural gas consumption does not appear to conform to the expected values when predicting into the future.

The predictions suggest Germany will remain economically stable in the near future, whilst still slowly phasing out greenhouse gas emitters as planned by the Energiewende. However these predictions could be misleading as the report has not analysed how the likelihood of an external shock due to geopolitical events, since these events, such as the COVID-19 pandemic, noticeably affect the time series that were analysed in the report.

The focus of this report has been to forecast the future of Germany's energy sector using time series models. The approach that was taken in this report is broadly applicable in the methodology to other countries provided there is enough time series data to make reasonable predictions into the future.

References

[1] C. E. Wire, "Germany's energiewende in brief," 2018, last accessed 22 February 2024. [Online]. Available: https://www.cleanenergywire.org/germanys-energiewende-brief

[2] Statistica, "Gross domestic product at current market prices of selected european countries in 2022," 2023, last accessed 22 February 2024. [Online]. Available: https://www.statista.com/statistics/685925/gdp-of-european-countries/